

This is a simple unbiased average of squared returns (since we are taking the trend term to be $C = 0$). The advantage of this approach is that the calculation is straightforward. The disadvantage is that the HMA assumes returns are independent and identically distributed with constant variance. However, this assumption does not seem to hold true over the period analyzed (see Figure 6.3a and Figure 6.3b). Since this model applies equal weightings to all historical data points it has been found to be slow to adapt to changing volatility regimes, such as the US debt crisis (Aug.–Sep. 2011) and during the financial crisis (Sep. 2008–Mar. 2009).

Exponential Weighted Moving Average (EWMA)

The exponential weighted moving average (EWMA) is computed as follows:

$$\hat{\sigma}_t^2 = (1 - \lambda)y_{t-1}^2 + \lambda\hat{\sigma}_{t-1}^2 \quad (6.17)$$

EWMA applies weights to the historical observations following an exponential smoothing process with parameter λ where $0 \leq \lambda \leq 1$. The value of the smoothing parameter is determined via maximum likelihood estimation (MLE). JP Morgan (1994) first introduced this model as part of their Risk Metrics offering.

The advantage of the EWMA is that it places more emphasis on the recent data observations. This allows the model to quickly update in a changing volatility environment. Additionally, its forecasts only require the previous period price change and the previous volatility forecast. We do not need to recalculate the forecast using a long history of price returns.

Arch Volatility Model

The ARCH volatility model was introduced by Engle (1982) and consists of the “p” previous returns. We can formulate it as follows:

$$\hat{\sigma}_t^2 = \omega + \sum_{i=1}^p \alpha_i r_{t-i}^2 \quad (6.18)$$

where, $\omega > 0$, $\alpha_1, \dots, \alpha_p \geq 0$, $\sum \alpha_i < 1$.

The parameters of the model are determined via ordinary least squares (OLS) regression analysis. The model differs from the HMA in that it does not apply the same weightings to all historical observations. To the extent that the more recent observations have a larger effect on current